

EE 516: Power Systems
Project Final
Due date: 8 pm, May 3, 2026

P1. A small load flow problem, is shown in Figure 11.11. Note the voltages, impedances and flows are all noted in per-unit. Using Matlab according to the MIT formulation:

- Write a Gauss–Seidel based program to solve for the voltage magnitudes and angles that result in the power flows described in the figure. Note that bus 4 here is the ‘swing bus’, which means its angle is the reference point.
- Note the voltage source at bus 3 is producing quite a lot of reactive power. How does this change if you inject $Q = 0.5$ per-unit into bus 5?
- What happens to the line flows if line 1 is removed?

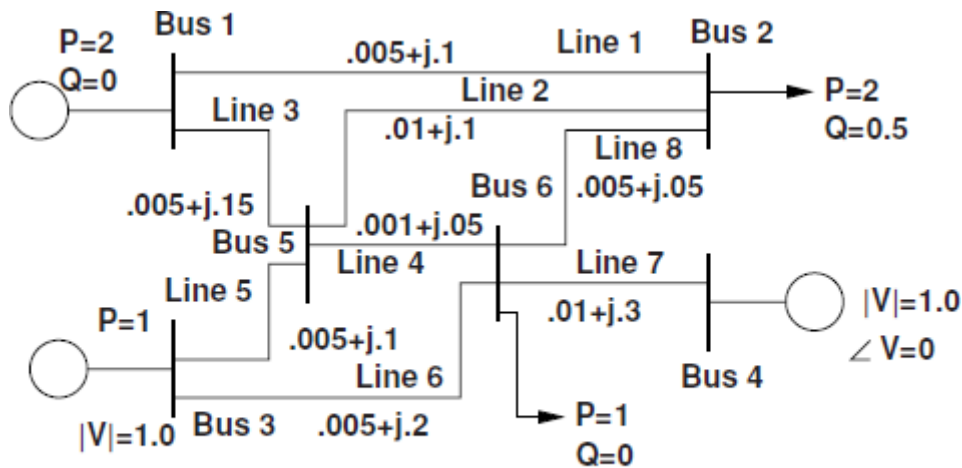


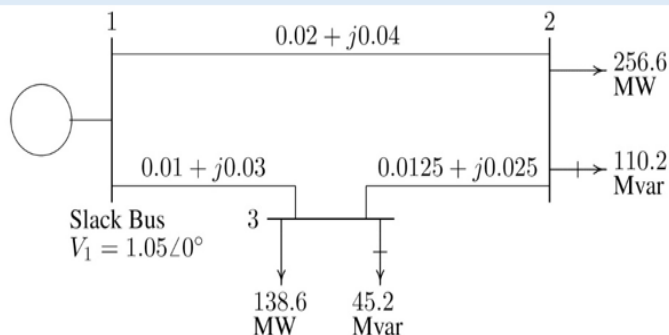
Figure 11.11 Small load flow example

P2. Solve example 7 using Matlab according to the MIT formulation.

Example 7

The one-line diagram of a simple three-bus power system is shown in the figure below. The generation is at bus 1. The $|V_1|$ at bus 1 is 1.05 pu. The scheduled loads on bus 2 and 3 are shown in the diagram. Line impedances are also shown as pu on a 100MVA base.

- Using the Gauss Seidel method, determine the phasor values of the voltage at the load buses 2 and 3 (P-Q buses) accurate to four decimal places.
- Find the slack bus real and reactive power.
- Determine the line flows and the line losses. Construct a power flow diagram showing the direction of line flow.



P3. Solve example 8 using Matlab according to the MIT methodology.



Example 8

Figure below shows the one-line diagram of a simple three-bus power system with generators at buses 1 and 3. The magnitude of the voltage at bus 1 is adjusted to 1.05 pu. Voltage magnitude at bus 3 is fixed at 1.04 pu with a real power generation of 200 MW. A load consisting of 400 MW and 250 MVAR is taken from bus 2. The base power is 100MVA. Obtain the power flow solution by using Gauss-Seidel method, then find the line flows and losses.

